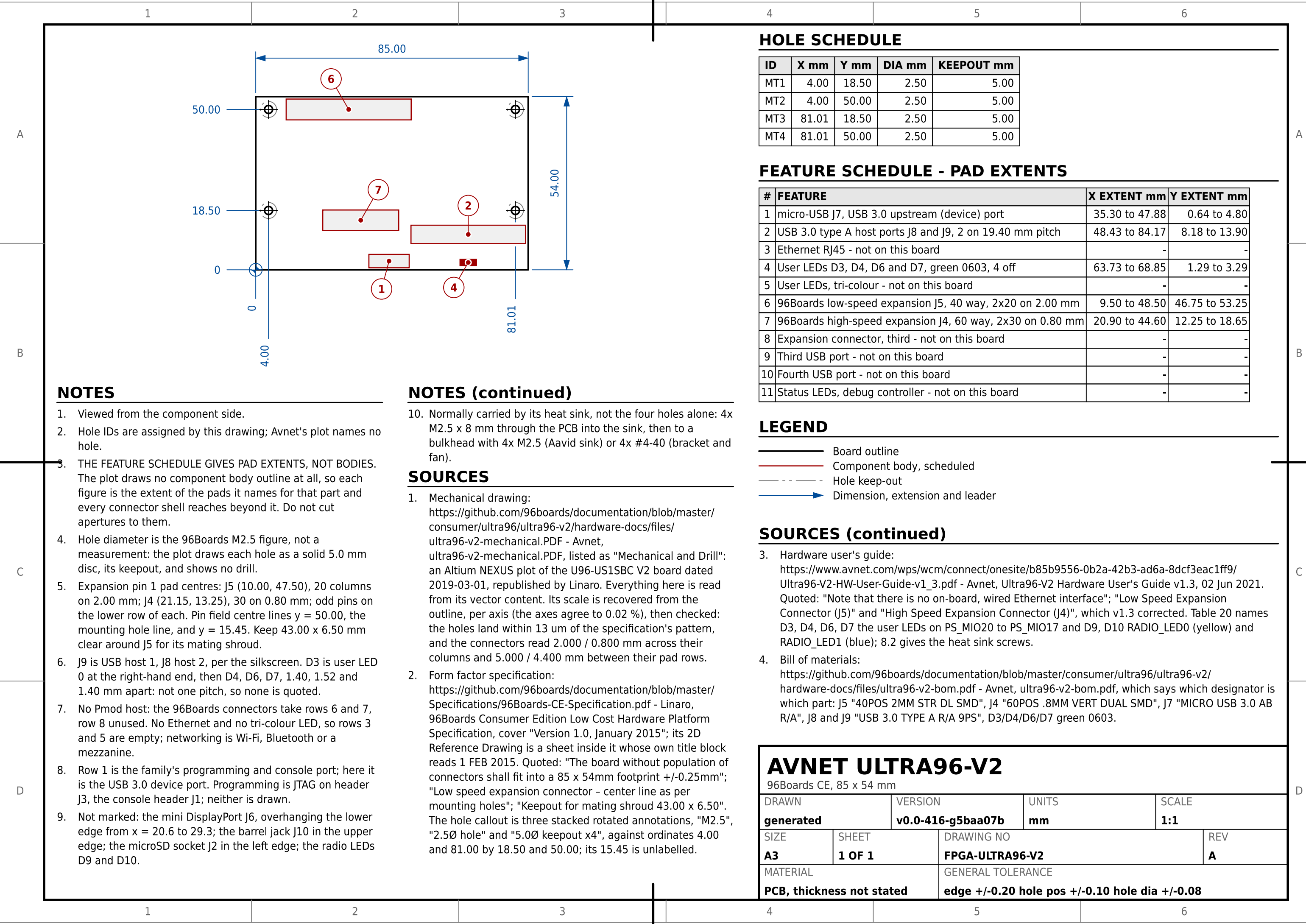




NOTES

- 1. Viewed from the component side.
- 2. Hole IDs are assigned by this drawing; Avnet's plot names no hole.
- 3. THE FEATURE SCHEDULE GIVES PAD EXTENTS, NOT BODIES. The plot draws no component body outline at all, so each figure is the extent of the pads it names for that part and every connector shell reaches beyond it. Do not cut apertures to them.
- 4. Hole diameter is the 96Boards M2.5 figure, not a measurement: the plot draws each hole as a solid 5.0 mm disc, its keepout, and shows no drill.
- 5. Expansion pin 1 pad centres: J5 (10.00, 47.50), 20 columns on 2.00 mm; J4 (21.15, 13.25), 30 on 0.80 mm; odd pins on the lower row of each. Pin field centre lines y = 50.00, the mounting hole line, and y = 15.45. Keep 43.00 x 6.50 mm clear around J5 for its mating shroud.
- 6. J9 is USB host 1, J8 host 2, per the silkscreen. D3 is user LED 0 at the right-hand end, then D4, D6, D7, 1.40, 1.52 and 1.40 mm apart: not one pitch, so none is quoted.
- 7. No Pmod host: the 96Boards connectors take rows 6 and 7, row 8 unused. No Ethernet and no tri-colour LED, so rows 3 and 5 are empty; networking is Wi-Fi, Bluetooth or a mezzanine.
- 8. Row 1 is the family's programming and console port; here it is the USB 3.0 device port. Programming is JTAG on header J3, the console header J1; neither is drawn.
- 9. Not marked: the mini DisplayPort J6, overhanging the lower edge from x = 20.6 to 29.3; the barrel jack J10 in the upper edge; the microSD socket J2 in the left edge; the radio LEDs D9 and D10.

NOTES (continued)

- 10. Normally carried by its heat sink, not the four holes alone: 4x M2.5 x 8 mm through the PCB into the sink, then to a bulkhead with 4x M2.5 (Aavid sink) or 4x #4-40 (bracket and fan).

SOURCES

- 1. Mechanical drawing: <https://github.com/96boards/documentation/blob/master/consumer/ultra96/ultra96-v2/hardware-docs/files/ultra96-v2-mechanical.PDF> - Avnet, ultra96-v2-mechanical.PDF, listed as "Mechanical and Drill": an Altium NEXUS plot of the U96-US1SBC V2 board dated 2019-03-01, republished by Linaro. Everything here is read from its vector content. Its scale is recovered from the outline, per axis (the axes agree to 0.02 %), then checked: the holes land within 13 um of the specification's pattern, and the connectors read 2.000 / 0.800 mm across their columns and 5.000 / 4.400 mm between their pad rows.
- 2. Form factor specification: <https://github.com/96boards/documentation/blob/master/Specifications/96Boards-CE-Specification.pdf> - Linaro, 96Boards Consumer Edition Low Cost Hardware Platform Specification, cover "Version 1.0, January 2015"; its 2D Reference Drawing is a sheet inside it whose own title block reads 1 FEB 2015. Quoted: "The board without population of connectors shall fit into a 85 x 54mm footprint +/-0.25mm"; "Low speed expansion connector - center line as per mounting holes"; "Keepout for mating shroud 43.00 x 6.50". The hole callout is three stacked rotated annotations, "M2.5", "2.5Ø hole" and "5.0Ø keepout x4", against ordinates 4.00 and 81.00 by 18.50 and 50.00; its 15.45 is unlabelled.

HOLE SCHEDULE

| ID  | X mm  | Y mm  | DIA mm | KEEPOUT mm |
|-----|-------|-------|--------|------------|
| MT1 | 4.00  | 18.50 | 2.50   | 5.00       |
| MT2 | 4.00  | 50.00 | 2.50   | 5.00       |
| MT3 | 81.01 | 18.50 | 2.50   | 5.00       |
| MT4 | 81.01 | 50.00 | 2.50   | 5.00       |

FEATURE SCHEDULE - PAD EXTENTS

| #  | FEATURE                                                   | X EXTENT mm    | Y EXTENT mm    |
|----|-----------------------------------------------------------|----------------|----------------|
| 1  | micro-USB J7, USB 3.0 upstream (device) port              | 35.30 to 47.88 | 0.64 to 4.80   |
| 2  | USB 3.0 type A host ports J8 and J9, 2 on 19.40 mm pitch  | 48.43 to 84.17 | 8.18 to 13.90  |
| 3  | Ethernet RJ45 - not on this board                         | -              | -              |
| 4  | User LEDs D3, D4, D6 and D7, green 0603, 4 off            | 63.73 to 68.85 | 1.29 to 3.29   |
| 5  | User LEDs, tri-colour - not on this board                 | -              | -              |
| 6  | 96Boards low-speed expansion J5, 40 way, 2x20 on 2.00 mm  | 9.50 to 48.50  | 46.75 to 53.25 |
| 7  | 96Boards high-speed expansion J4, 60 way, 2x30 on 0.80 mm | 20.90 to 44.60 | 12.25 to 18.65 |
| 8  | Expansion connector, third - not on this board            | -              | -              |
| 9  | Third USB port - not on this board                        | -              | -              |
| 10 | Fourth USB port - not on this board                       | -              | -              |
| 11 | Status LEDs, debug controller - not on this board         | -              | -              |

LEGEND

- Board outline
- Component body, scheduled
- Hole keep-out
- Dimension, extension and leader

SOURCES (continued)

- 3. Hardware user's guide: [https://www.avnet.com/wps/wcm/connect/onesite/b85b9556-0b2a-42b3-ad6a-8dcf3eac1ff9/Ultra96-V2-HW-User-Guide-v1\\_3.pdf](https://www.avnet.com/wps/wcm/connect/onesite/b85b9556-0b2a-42b3-ad6a-8dcf3eac1ff9/Ultra96-V2-HW-User-Guide-v1_3.pdf) - Avnet, Ultra96-V2 Hardware User's Guide v1.3, 02 Jun 2021. Quoted: "Note that there is no on-board, wired Ethernet interface"; "Low Speed Expansion Connector (J5)" and "High Speed Expansion Connector (J4)", which v1.3 corrected. Table 20 names D3, D4, D6, D7 the user LEDs on PS\_MIO20 to PS\_MIO17 and D9, D10 RADIO\_LED0 (yellow) and RADIO\_LED1 (blue); 8.2 gives the heat sink screws.
- 4. Bill of materials: <https://github.com/96boards/documentation/blob/master/consumer/ultra96/ultra96-v2/hardware-docs/files/ultra96-v2-bom.pdf> - Avnet, ultra96-v2-bom.pdf, which says which designator is which part: J5 "40POS 2MM STR DL SMD", J4 "60POS .8MM VERT DUAL SMD", J7 "MICRO USB 3.0 AB R/A", J8 and J9 "USB 3.0 TYPE A R/A 9PS", D3/D4/D6/D7 green 0603.

AVNET ULTRA96-V2

96Boards CE, 85 x 54 mm

|                                              |                        |                                                                            |                    |                     |                 |
|----------------------------------------------|------------------------|----------------------------------------------------------------------------|--------------------|---------------------|-----------------|
| DRAWN<br><b>generated</b>                    |                        | VERSION<br><b>v0.0-416-g5baa07b</b>                                        | UNITS<br><b>mm</b> | SCALE<br><b>1:1</b> |                 |
| SIZE<br><b>A3</b>                            | SHEET<br><b>1 OF 1</b> | DRAWING NO<br><b>FPGA-ULTRA96-V2</b>                                       |                    |                     | REV<br><b>A</b> |
| MATERIAL<br><b>PCB, thickness not stated</b> |                        | GENERAL TOLERANCE<br><b>edge +/-0.20 hole pos +/-0.10 hole dia +/-0.08</b> |                    |                     |                 |